# PRACTICAL: 10

## AIM:

A mid-sized company plans to enhance network security by deploying a robust firewall solution. The IT security team is tasked with studying and testing various firewall software options to determine the most suitable one based on functionality, ease of use, and performance. To evaluate and compare different firewall software solutions for securing network traffic and preventing unauthorised access in a simulated enterprise environment.

## THEORY:

**Types of Firewalls:**

A firewall is a network security system that protects a computer network from unauthorized access and cyber attacks. It monitors incoming and outgoing network traffic and allows or blocks data packets based on predefined security rules. Firewalls act as a barrier between a trusted internal network and an untrusted external network such as the Internet. They help organizations maintain security by preventing hackers, malware, and unauthorized users from accessing sensitive data.

There are different types of firewalls used in network security. Some of the commonly used firewalls are explained below.

1. **Packet Filtering Firewall**

   A packet filtering firewall is the simplest and oldest type of firewall. It checks the header of each data packet and decides whether to allow or block it based on rules such as source IP address, destination IP address, port number, and protocol.

   This type of firewall works at the network layer of the OSI model. It is fast and requires fewer system resources. However, it does not inspect the content of the packet, so it cannot detect advanced attacks.

2. **Stateful Inspection Firewall**

   A stateful inspection firewall improves upon packet filtering by keeping track of the state of active network connections. It monitors ongoing sessions and determines whether a packet belongs to a valid and trusted connection.

   Because it understands the state of network traffic, it provides better security than simple packet filtering firewalls. It is widely used in modern network security systems.

3. **Proxy Firewall**

   A proxy firewall, also known as an application-level gateway, acts as an intermediary between the user and the destination server. Instead of direct communication, the firewall receives the request, analyzes it, and then forwards it to the server.

This firewall works at the application layer of the OSI model and can inspect application-level data such as HTTP and FTP traffic. It provides a higher level of security but may slow down network performance.

### 4. Next-Generation Firewall (NGFW)

A next-generation firewall (NGFW) is an advanced firewall that combines traditional firewall functions with modern security technologies. It provides additional security features such as intrusion prevention systems (IPS), deep packet inspection, malware detection, and application awareness.

NGFWs are widely used in enterprise networks because they offer strong protection against modern cyber threats.

### 5. Web Application Firewall (WAF)

A web application firewall (WAF) is designed specifically to protect web applications from attacks. It filters and monitors HTTP and HTTPS traffic between users and web servers.

WAF protects websites from common web-based attacks such as SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF). It is commonly used to secure websites, APIs, and online services.

## SCREENSHOTS:

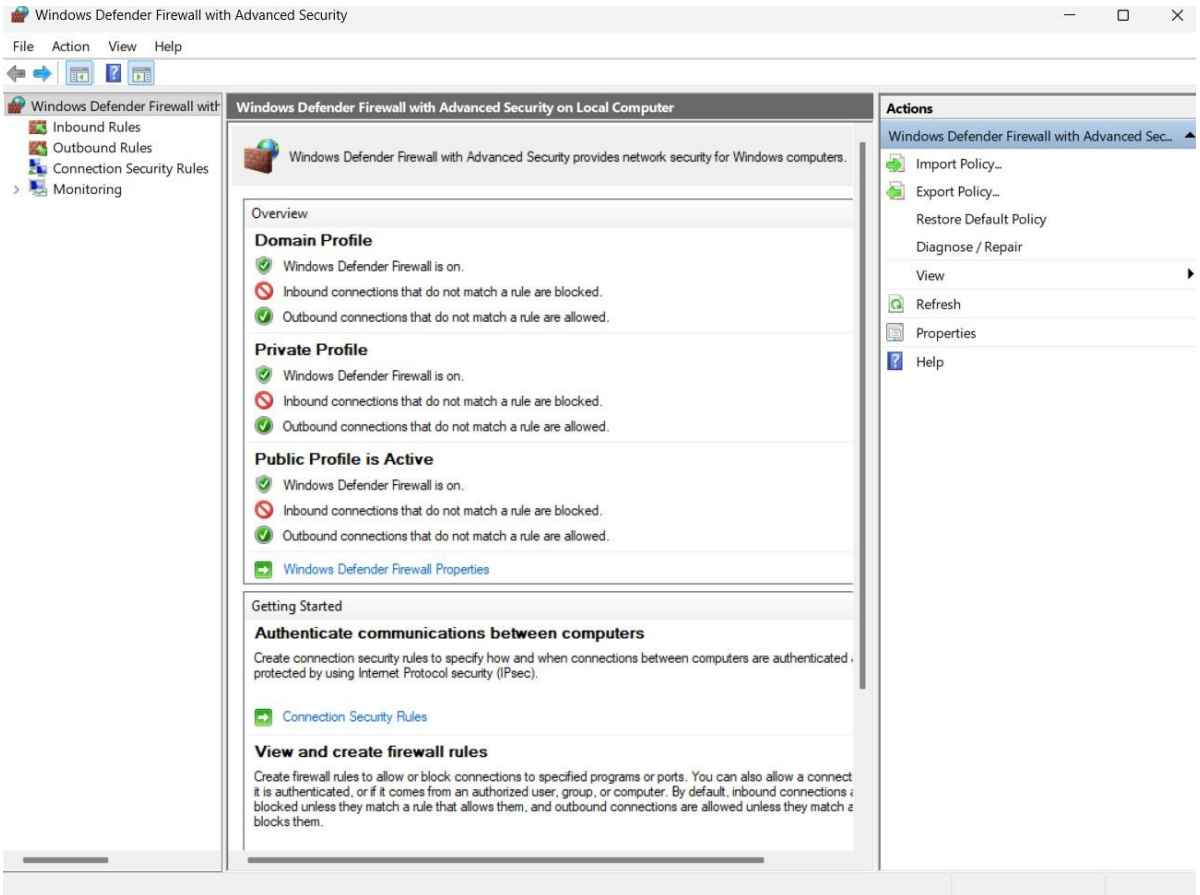


*Figure 10. 1 : Windows Defender Firewall*

TASK 1) Block Internet Access

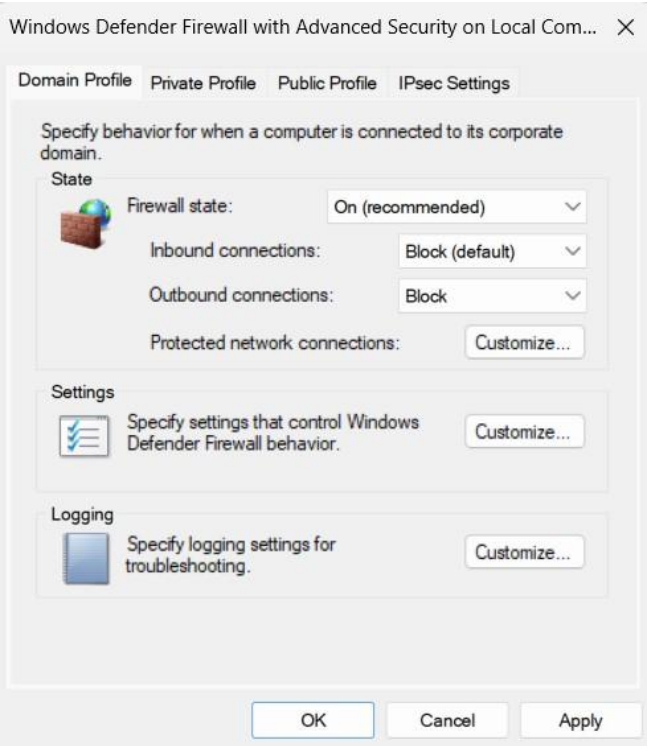


*Figure 10. 2 : Outbound rule edit*

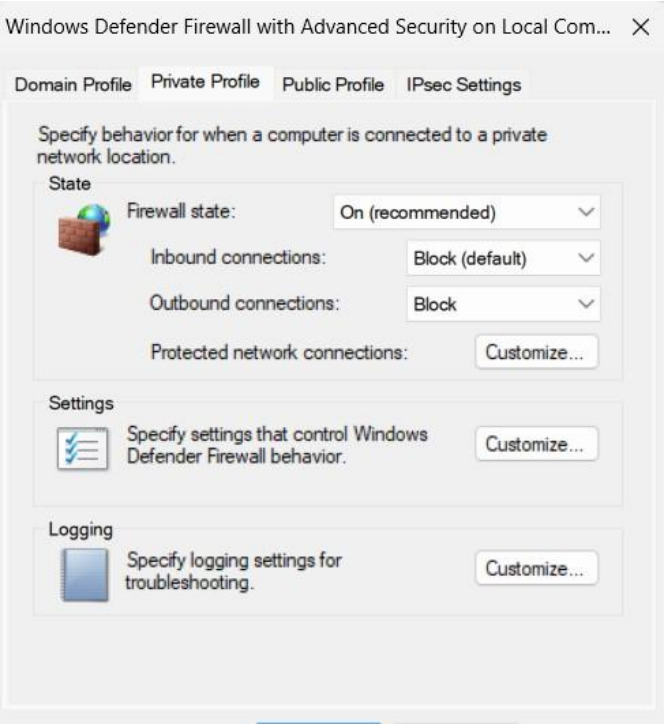


*Figure 10. 3: Outbound rule edit*

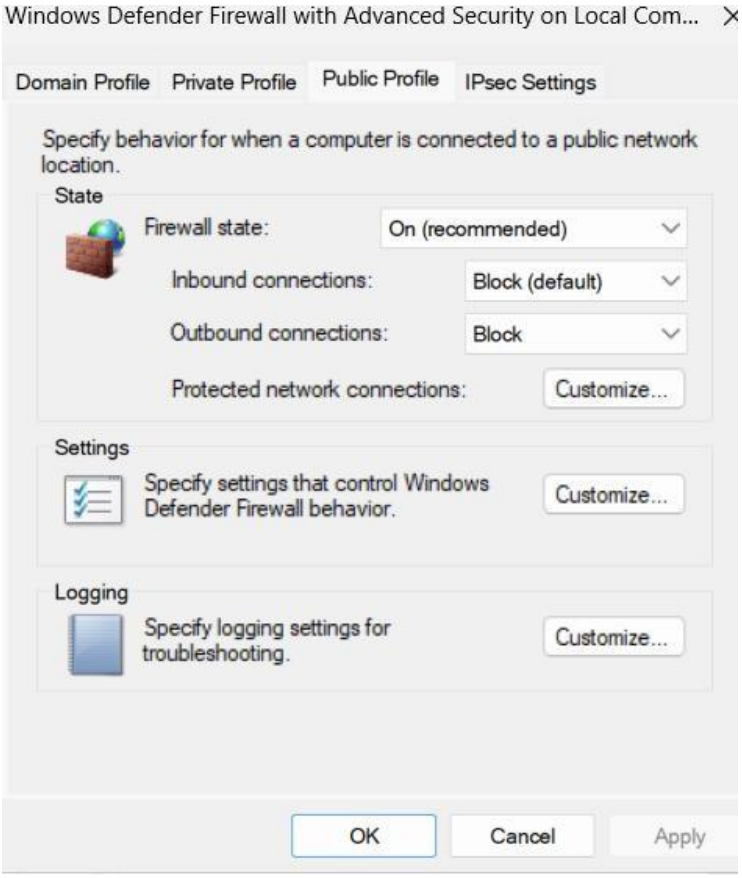


*Figure 10. 4 : Outbound rule edit*

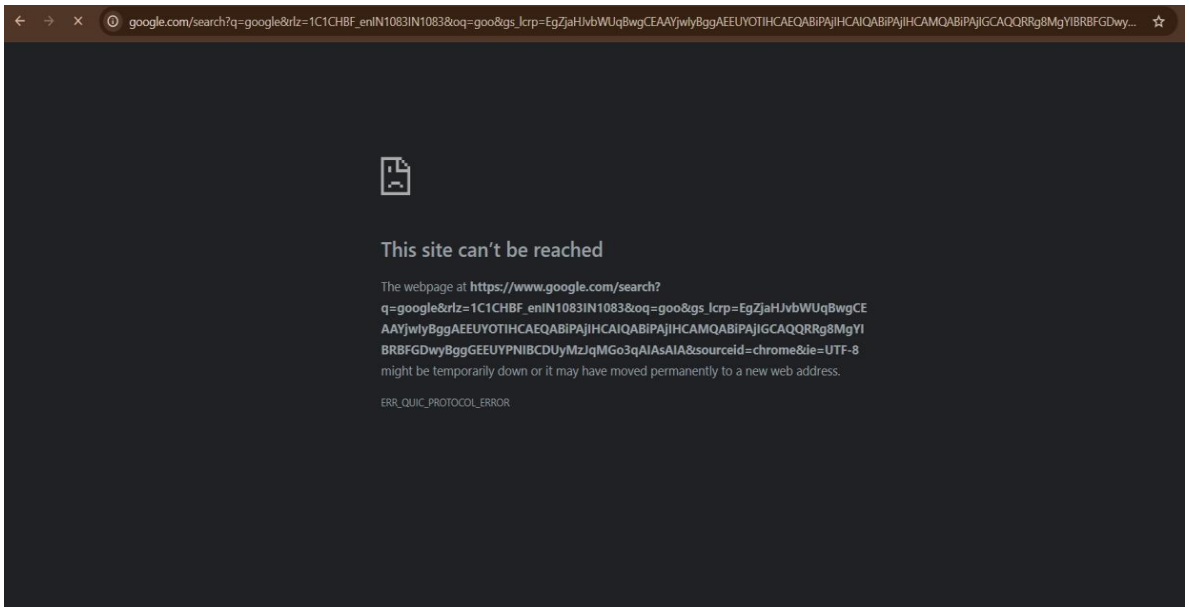


*Figure 10. 5 : site not reachable*

TASK 2) Block Browser app using firewall

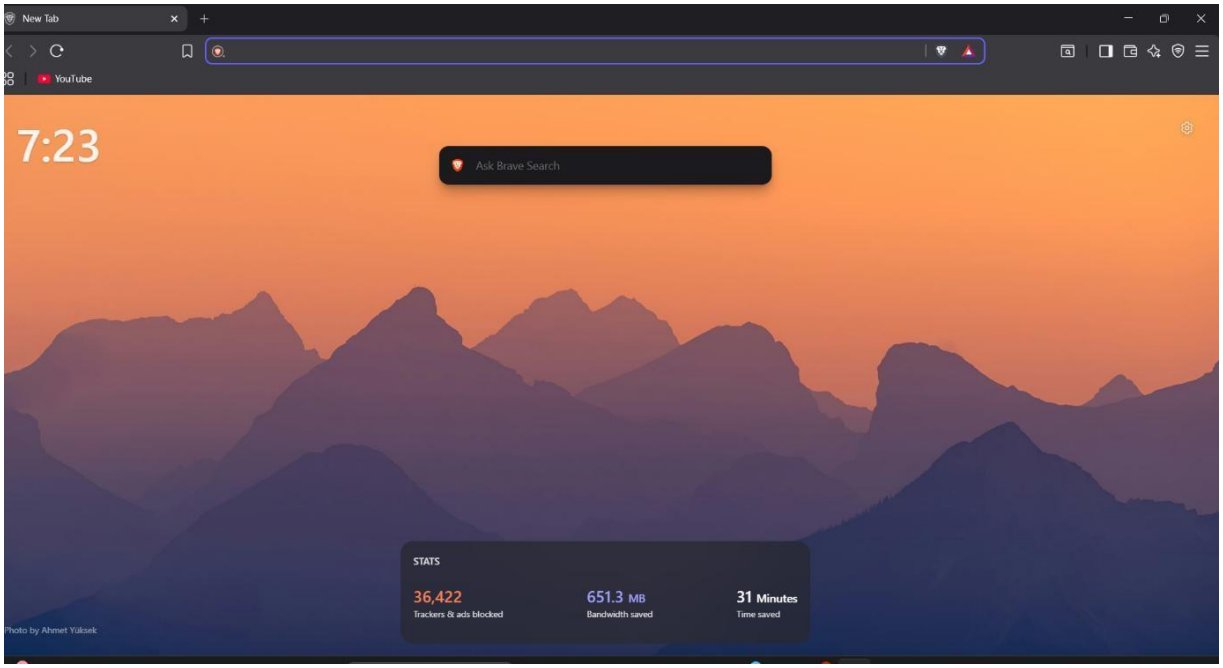


*Figure 10. 6 : running browser*

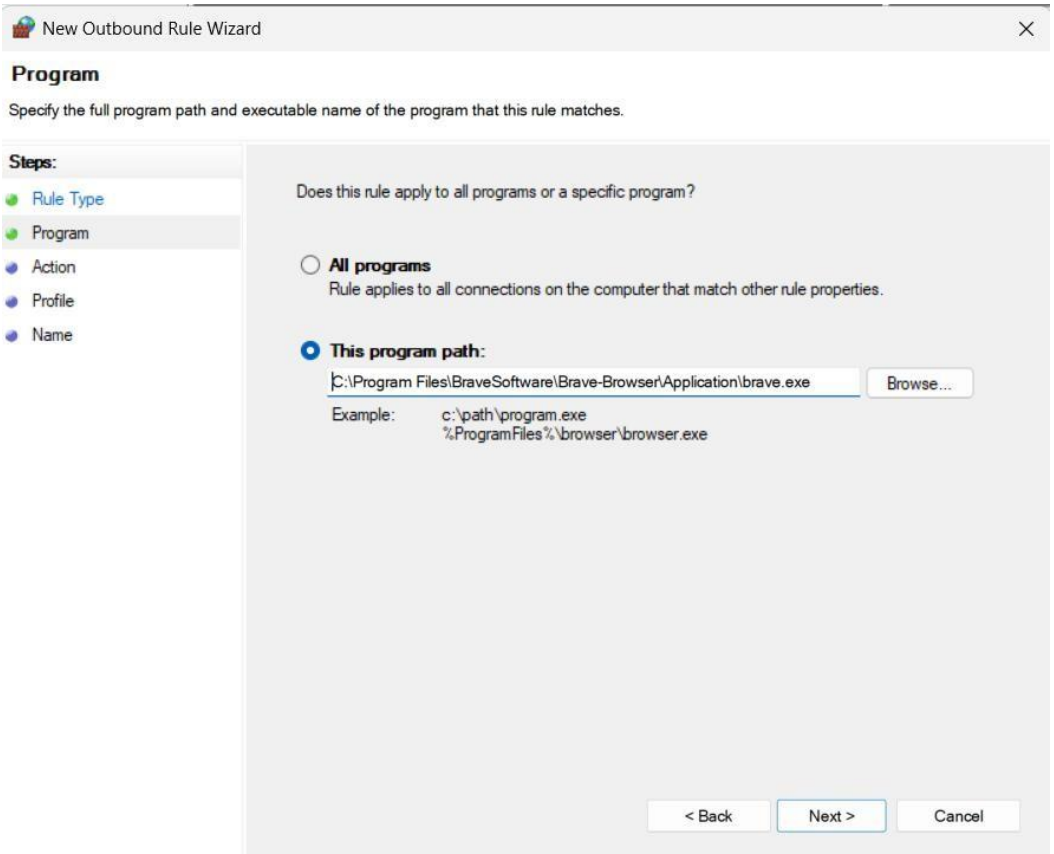


*Figure 10. 7 : Set program and add .exe path*

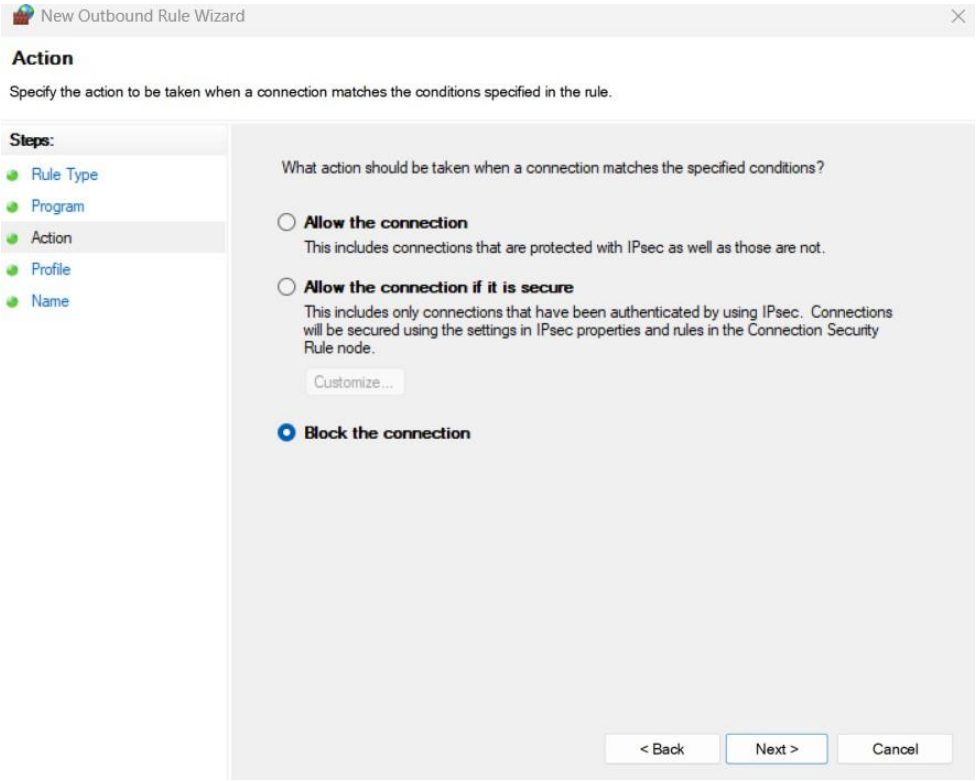


*Figure 10. 8 : Block the Connection*

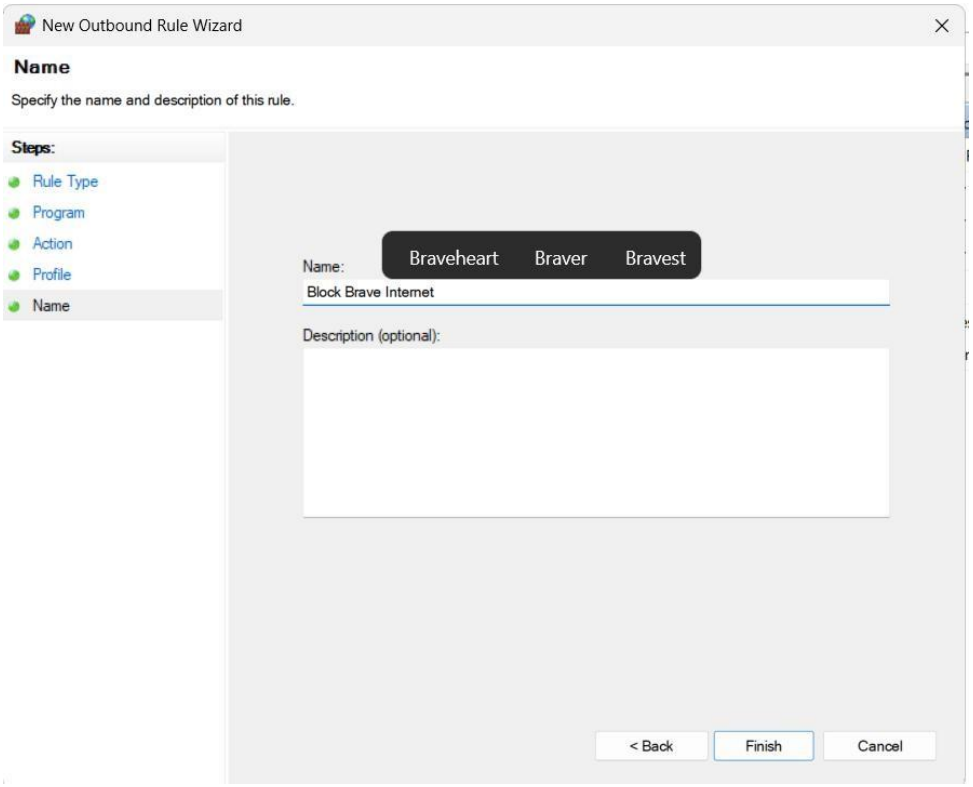


*Figure 10. 9 : Name the firewall*

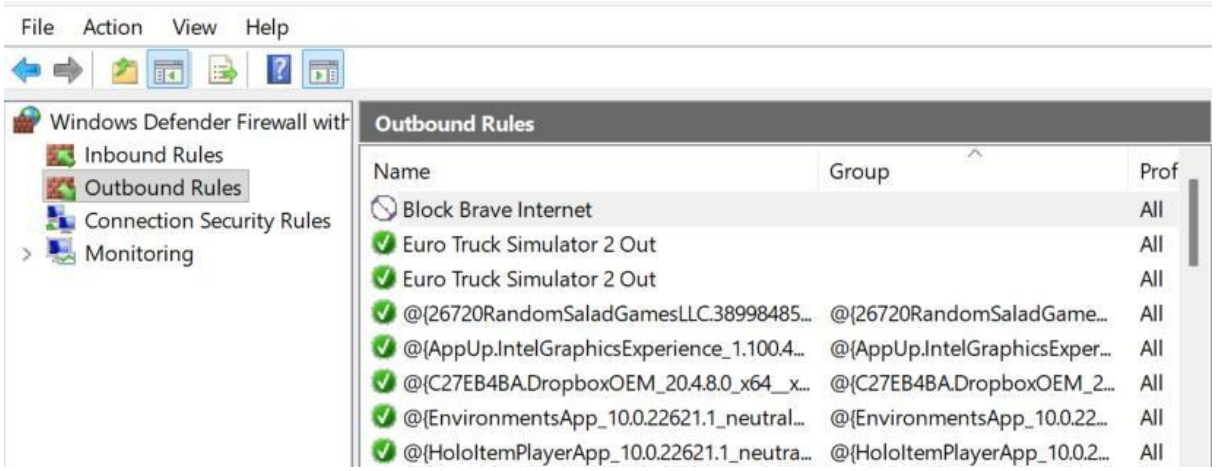


*Figure 10. 10 : Outbound rule*

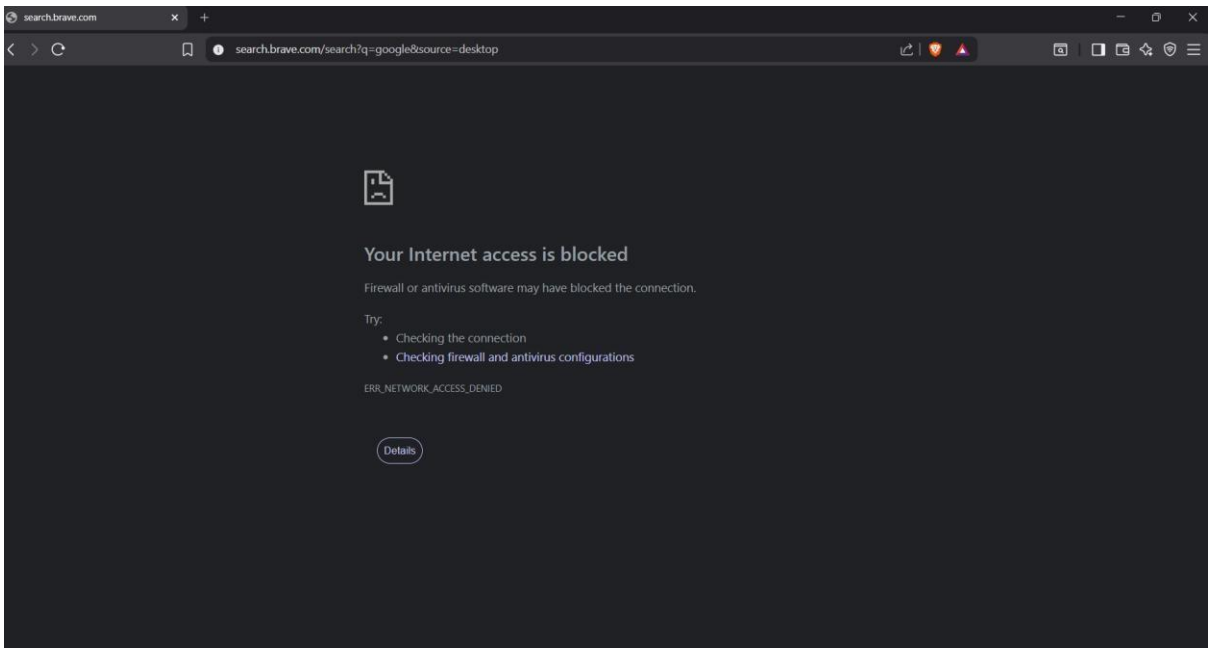


*Figure 10. 11 : access is blocked*

TASK 3) Ping block using firewall

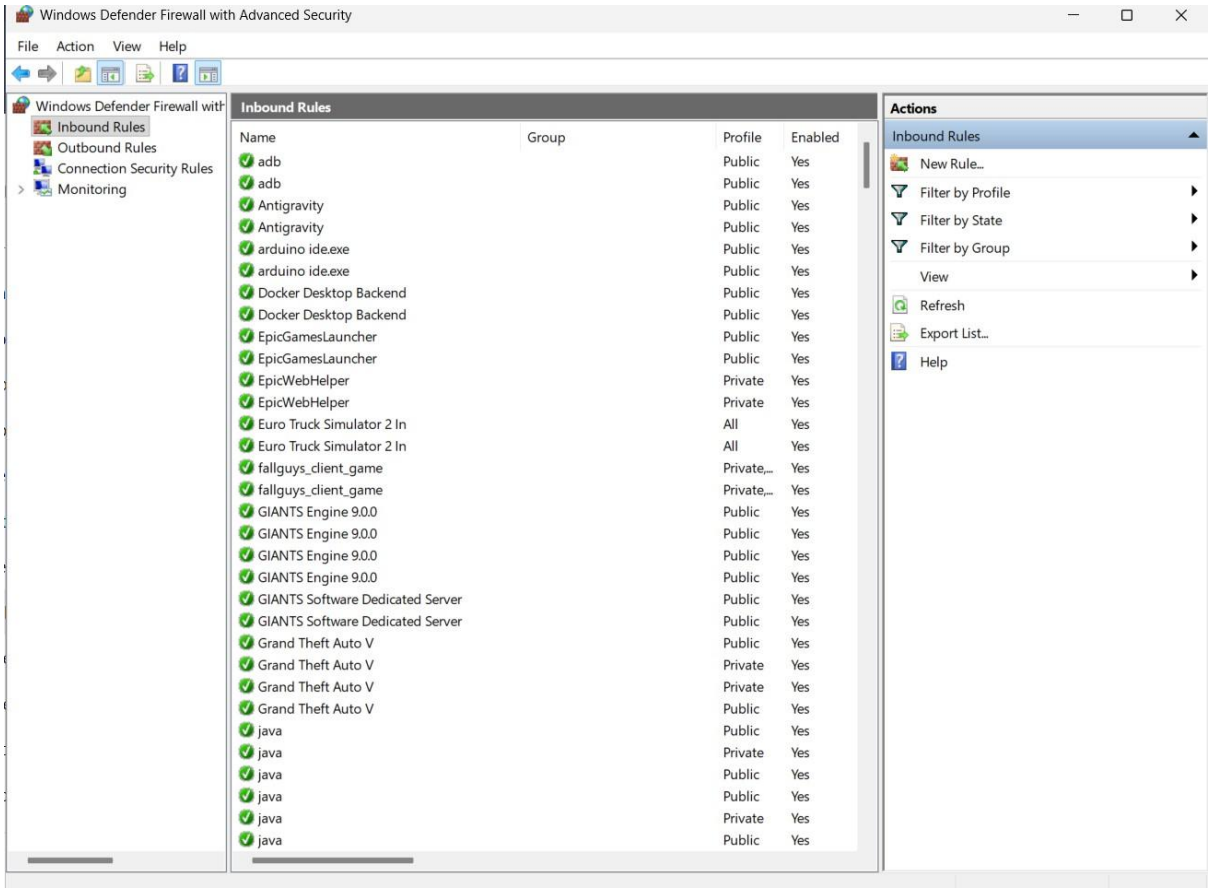


*Figure 10. 12 : inbound rule configure*

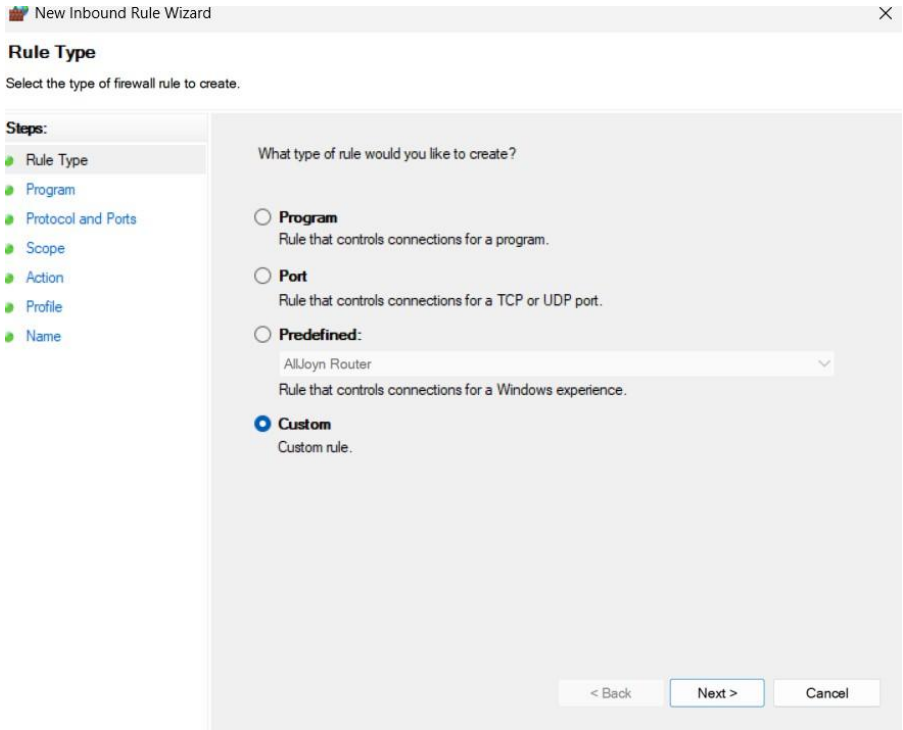


*Figure 10. 13 : select custom*

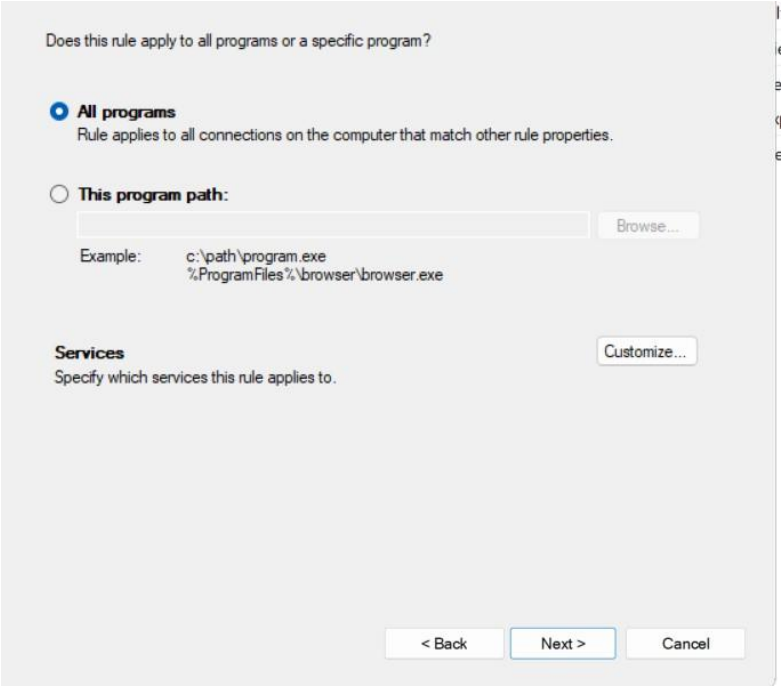


*Figure 10. 14 : Select all programs*

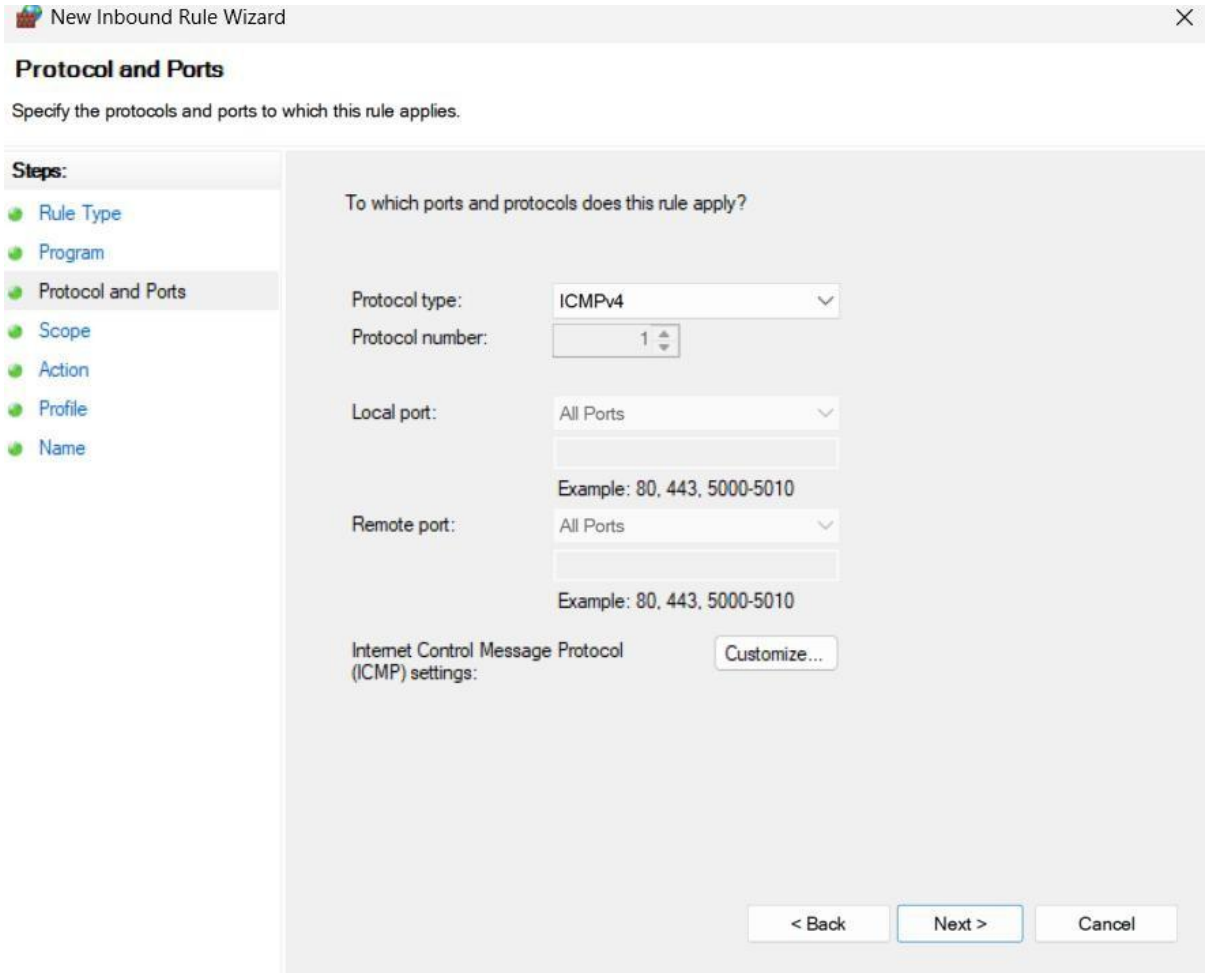


*Figure 10. 15 : select ICMPv4*

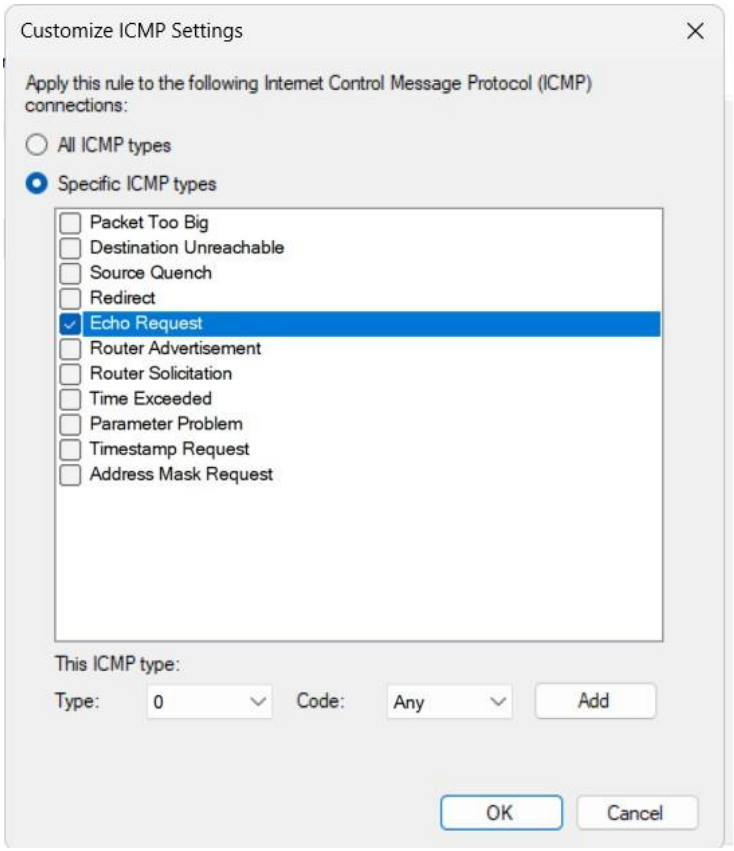


*Figure 10. 16 : select Echo req*

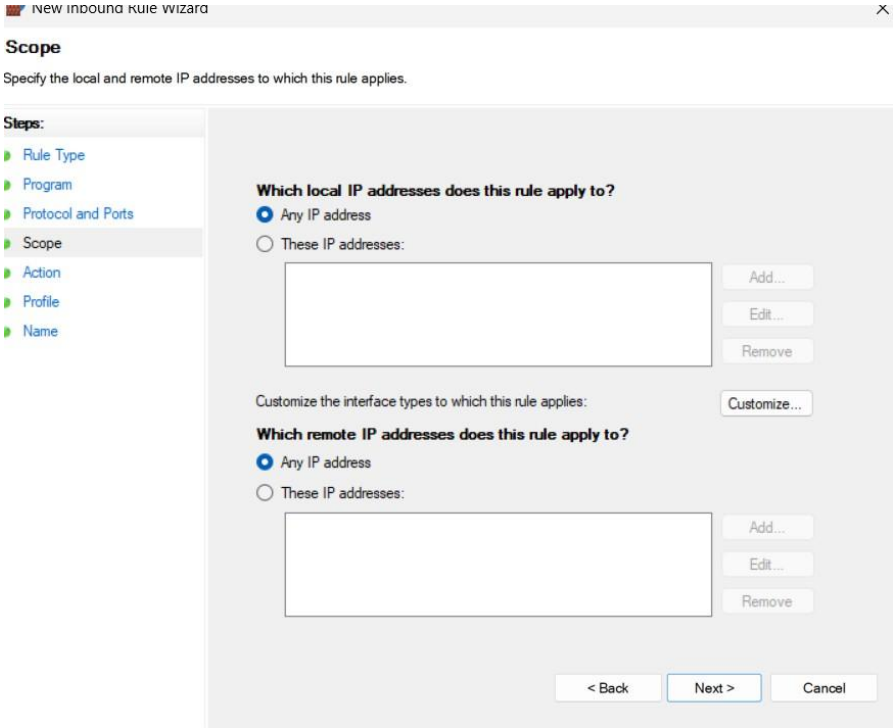


*Figure 10. 17 : Select Any IP*

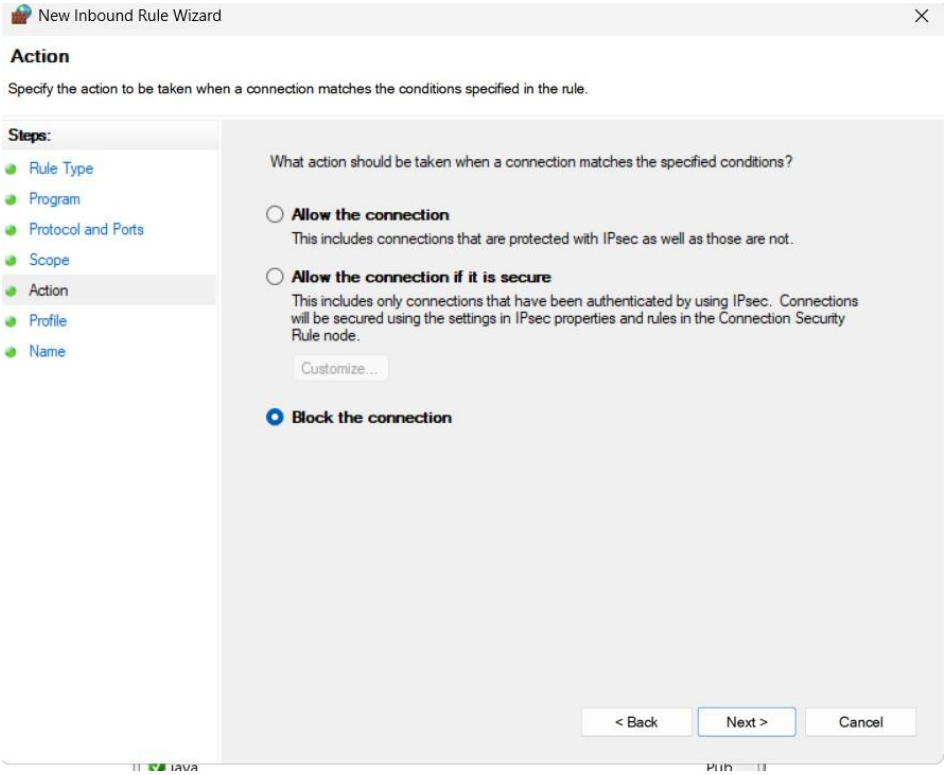


*Figure 10. 18 : Block connection*

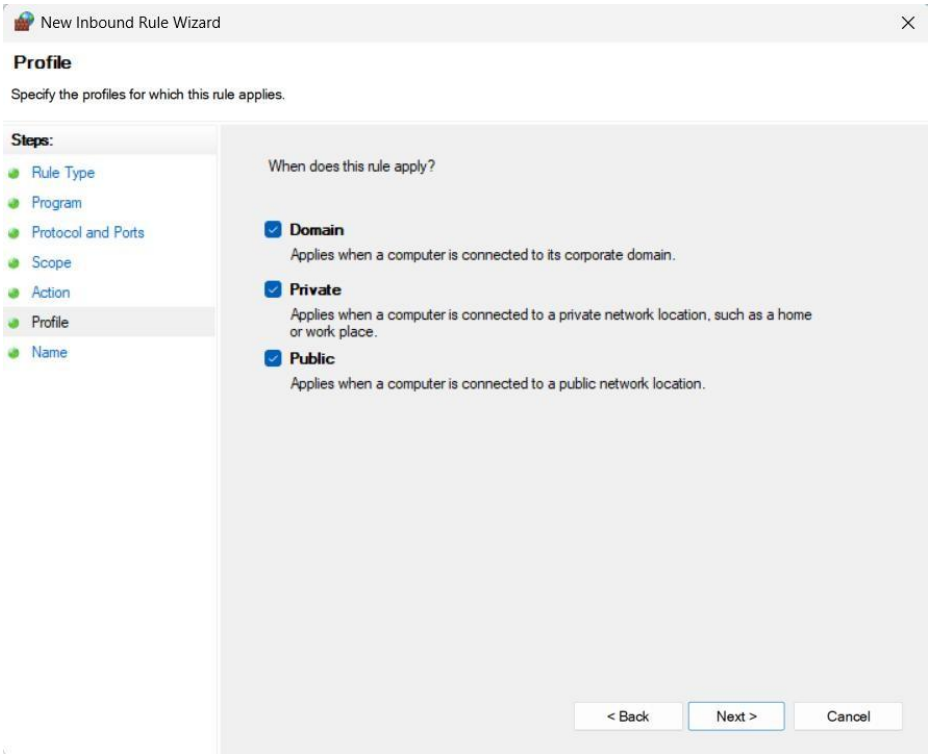


*Figure 10. 19 : select all*

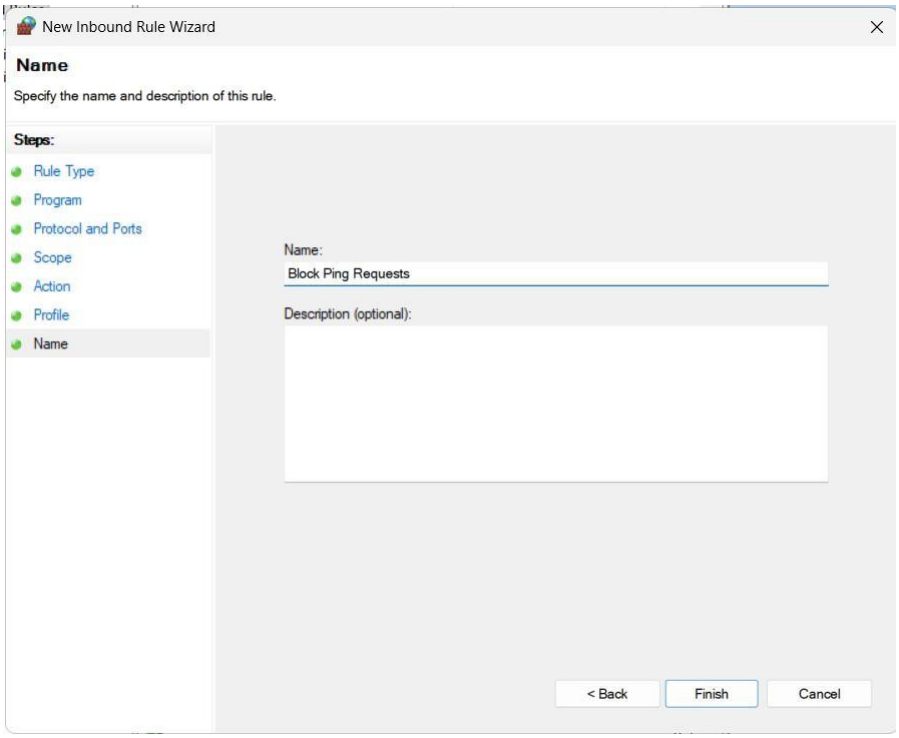


*Figure 10. 20 : Name firewall*

## LEARNING OUTCOME:

Firewalls play a very important role in network security. They help protect systems and networks from unauthorized access, cyber attacks, and malicious traffic. Different types of firewalls provide different levels of protection, and organizations often use multiple firewall technologies together to improve their overall security.

## REFERENCES:

**Windows Defender Firewall - https://www.youtube.com/watch?v=pP7_nFBNR-M**